Equivalence of Locally Connected Definitions

October 21, 2026

Point-Set Topology

Definition 1: Locally Connected - Def 1

A topological space X is said to be locally connected if for each $x \in X$ and for each open neighbourhood U containing x, there exists a connected neighbourhood V such that $x \in V \subseteq U$.

Definition 2: Locally Connected - Def 2

A topological space X is said to be locally connected if for each $x \in X$ and for each open neighbourhood U containing x, there exists a connected *open* neighbourhood V such that $x \in V \subseteq U$.

Problem 1: Equivalence

Are Def 1 and Def 2 equivalent?

Solution Roadmap

Def 2 definitely implies Def 1. The interesting part is proving the reverse.

Let Def 1 hold.

1. **Step 1:** Show each connected component is open.
2. **Step 2:** Each open set U is a locally connected space (LCP).
3. **Step 3:** By Step 1, each connected component of U is open.
4. **Step 4:** Now for each x and for each open neighbourhood U of x, take V as the connected component of x in U. It is open since it is open in U.